

Multiple Choice

1. **Answer:** A

Options: A) Temporary inconsistencies among views of a file by different machines can result; B) The file system is likely to be corrupted when a computer crashes; C) A much higher amount of network traffic results; D) Caching makes file migration impossible

Note: Correct option: A - Temporary inconsistencies among views of a file by different machines can result

2. **Answer:** D

Options: A) $O(\log N)$; B) $O(N \log N)$; C) $O(\log N) + O(1)$; D) $O((\log N)^2)$

Note: Correct option: D - $O((\log N)^2)$

3. **Answer:** C

Options: A) C, Ruby; B) Python, Ruby; C) C, C++; D) Tcl, C#

Note: Correct option: C - C, C++

4. **Answer:** A

Options: A) Local variables; B) Program code; C) Dynamically linked libraries; D) Global variables

Note: Correct option: A - Local variables

5. **Answer:** B

Options: A) Finding a longest simple cycle in G; B) Finding a shortest cycle in G; C) Finding ALL spanning trees of G; D) Finding a largest clique in G

Note: Correct option: B - Finding a shortest cycle in G

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Only once'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It can never be proven.'

Long Form Response

11. **Answer:** `def remove_datatype(test_tuple, data_type): | return (res)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def num_position(text): | return m.start()`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key